

Department of Computer Science and Engineering
National Institute of Technology Calicut
Tentative Course Plan- Monsoon 2025
CS4023D Artificial Intelligence
(Seventh Semester B. Tech CSE)

Lecture:

Lecture Hours : E1, E2 Slots

Online Lectures/Resources through : NITC Eduserver (<https://eduserver.nitc.ac.in/course/view.php?id=8600>)

Enrollment Key : 6ycvrs



Instructors:

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Course Outcomes

CO1: Apply State-space search strategies for problem solving and implement it using LISP.
CO2: Use heuristic functions in search strategies and Games.
CO3: Apply various knowledge representation mechanisms.
CO4: Identify and choose the appropriate machine learning approach for solving various problems.

Evaluation Scheme:

Mid-Term Exam: 20

Assignments/Quizzes/Lessons/Tutorials : 40

Assignment /Lesson/Quiz : 4* 10 = 40

End Semester Exam : 40

Weekly Plan:

Week	Faculty	Activity	Topic	Details
1	LA	Lectures	Introduction	Artificial Intelligence: Introduction, History and Applications; Intelligent Agents
2	LA	Lectures	State Space	Solving problems by Searching: Structures and Strategies for state space search- Data driven and goal driven search, Uninformed Search strategies,
3	LA	Lectures Assignment-1	Informed Search	Informed(Heuristic) Search Strategies, Heuristic Functions, Local Search Algorithms and Optimization Problems, Searching with Nondeterministic actions,
4	LA SW	Lectures Quiz-1	Two Agent games, Introduction to Machine Learning	Constraint satisfaction, Using heuristics in games- Minimax Search, Alpha Beta Procedure, Stochastic Games. Supervised Learning, Decision Tree, optimization, linear regression
5	SW	Lectures Assignment-2	Machine learning	Error and noise & Training and testing, Logistic regression, Bias, variance tradeoff, Overfitting, SVM, K-Means
6	SW	Lectures	Probabilistic model	Statistical Learning, Hidden Variables, HMM, CRF, Maximum entropy Model
7	SW	Lectures Quiz-2	Reinforcement Learning	Rewards, passive reinforcement Learning, Active reinforcement Learning, policy Search, application of Reinforcement Learning
8			Midterm	
9			Midterm	
9	CMC	Lectures	Knowledge representation	Knowledge based agents, Propositional calculus, First-Order Logic (Predicate Calculus), Inference in First-Order Logic, Forward and

				Backward chaining
10	CMC	Lectures (Assignment-3)	Answer Extraction	Theorem proving by Resolution, Answer Extraction
11	CMC	Lectures	AI Representational Schemes	Semantic Nets, Conceptual Dependency, Scripts, Frames
12	CMC JP	Lectures (Quiz-3)	Planning Introduction to NLP & Neural Networks	Planning and acting in the real world Introduction to NLP & Language Models, Neural Networks
13	JP	Lectures	Deep Learning: Basic Models using Python	Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN),
14	JP	Lectures	Transformer Model	Transformer Architecture, Attention Mechanism
15	JP	Lectures & Quiz	Concluding Remarks	Use cases, Limit & Ethics of AI
16			Endsem	
17			Endsem	

Grading Policy:

- Grading would be relative, but 30% marks are mandatory for a pass grade
- The institute regulations with respect to attendance would be strictly adhered to
- Copying in exams and other breaches of Academic Integrity (refer <http://cse.nitc.ac.in/sites/default/files/Academic-Integrity.pdf>) will be very strictly penalized.

References:

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 2/e, Pearson Education, 2002.
2. G. F. Luger, *Artificial Intelligence- Structures and Strategies for Complex Problem Solving*, 4/e, 2002, Pearson Education.